

GPU Accelerated Reduced Order Modelling and Analysis of NACA0012 Airfoil

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Tonal noise emitted from the trailing edge of an airfoil is considered using modal techniques to investigate secondary quadrupole and dipole tones. We examine the origin of quadrupole and dipole sound generated from 2-D unsteady flow over a NACA0012 airfoil. In this paper, we consider two flow configurations at Mach numbers of $M_\infty = 0.3$ and 0.5 that lead to different acoustic characteristics: the later has a significant high-frequency quadrupole noise source. We employ Dynamic mode decomposition (DMD) to reveal the quadrupole sound and how it varies across different Mach numbers. It is found that the quadrupole sound is caused by vortex shedding in the vicinity of the trailing edge. The complex vortex interaction between both sides of the airfoil strengthens the quadrupole source.

I. Nomenclature

<i>FOM</i>	=	Full Order Model
<i>ROM</i>	=	Reduced Order Model
<i>St</i>	=	Strouhal Number
<i>POD</i>	=	Proper Orthogonal Decomposition
<i>DMD</i>	=	Dynamic Mode Decomposition
<i>tls - dmd</i>	=	Total Least Squares Dynamic Mode Decomposition
<i>PSD</i>	=	Power Spectral Density

II. Introduction

Trailing-edge (TE) noise remains a critical concern in the design of modern aircraft and Unmanned Air Vehicles (UAVs). Extensive prior research has focused on the dominant low-frequency dipole sound source, but recent studies, such as the work by Kojima et al. (2023) [1], emphasize the importance of analyzing higher-frequency secondary tones associated with a distinct quadrupole source. Despite these insights, the specific physical mechanism and flow conditions that govern the generation and strength of these quadrupole tones, especially in the regime of moderate to high subsonic Mach numbers, require further investigation. This study examines the origin of these quadrupole and dipole sound generated from unsteady flow over a NACA0012 aerofoil, specifically focusing on flow configurations at $M_\infty = 0.3$ and 0.5 . To efficiently process high-fidelity flow data, this work employs advanced modal analysis techniques, particularly Dynamic Mode Decomposition (DMD), within a GPU Accelerated Reduced Order Modelling (ROM) framework. The results confirm that the generation of quadrupole sound is intrinsically linked to vortex shedding in the vicinity of the trailing edge. Crucially, the complex interaction between the vortices shed from both the suction and pressure sides of the aerofoil acts to substantially strengthen the overall quadrupole noise source, particularly in the higher Mach number case ($M_\infty = 0.5$). This finding provides essential insights into the underlying fluid-acoustic coupling responsible for high-frequency secondary tones.

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III. Governing Equation and Numerical Setup

A. Governing Equation

Unsteady three-dimensional compressible Navier-Stokes equations governing the fluid flow are transformed into the generalized curvilinear coordinate system. The following are the resulting governing equations in conservative form.

$$\frac{\partial}{\partial t} \left(\frac{U}{J} \right) = \frac{\partial}{\partial \xi} \left(\frac{1}{Re} F_v - F_I \right) + \frac{\partial}{\partial \eta} \left(\frac{1}{Re} G_v - G_I \right) + \frac{\partial}{\partial \zeta} \left(\frac{1}{Re} H_v - H_I \right) \quad (1)$$

Here t is the time, ξ, η, ζ are the transformed coordinates, U is the dependent variable vector, J -Jacobian of the transformation matrix, F_I, G_I, H_I are the inviscid flux vectors and F_v, G_v, H_v are the viscous flux vectors. Inviscid flux vectors correspond to convection whereas Viscous flux vectors correspond to Diffusion phenomena. Following Visbal and Gaitonde[2], these vectors are given by:

$$U_{con} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E_t \end{bmatrix}, \quad F_I = \frac{1}{J} \begin{bmatrix} \rho U \\ \rho u U + \xi_x p \\ \rho v U + \xi_y p \\ \rho w U + \xi_z p \\ \rho E_t U + p \hat{U} \end{bmatrix}, \quad G_I = \frac{1}{J} \begin{bmatrix} \rho V \\ \rho u V + \eta_x p \\ \rho v V + \eta_y p \\ \rho w V + \eta_z p \\ \rho E_t V + p \hat{V} \end{bmatrix}, \quad H_I = \frac{1}{J} \begin{bmatrix} \rho W \\ \rho u W + \zeta_x p \\ \rho v W + \zeta_y p \\ \rho w W + \zeta_z p \\ \rho E_t W + p \hat{W} \end{bmatrix},$$

$$F_v = \frac{1}{J} \begin{bmatrix} 0 \\ \xi_{x_i} \tau_{i1} \\ \xi_{x_i} \tau_{i2} \\ \xi_{x_i} \tau_{i3} \\ \xi_{x_i} b_i \end{bmatrix}, \quad G_v = \frac{1}{J} \begin{bmatrix} 0 \\ \eta_{x_i} \tau_{i1} \\ \eta_{x_i} \tau_{i2} \\ \eta_{x_i} \tau_{i3} \\ \eta_{x_i} b_i \end{bmatrix}, \quad H_v = \frac{1}{J} \begin{bmatrix} 0 \\ \zeta_{x_i} \tau_{i1} \\ \zeta_{x_i} \tau_{i2} \\ \zeta_{x_i} \tau_{i3} \\ \zeta_{x_i} b_i \end{bmatrix}$$

U, V and W are contravariant components of velocity, defined by:

$$U = \xi_x u + \xi_y v + \xi_z w$$

$$V = \eta_x u + \eta_y v + \eta_z w$$

$$W = \zeta_x u + \zeta_y v + \zeta_z w$$

$$E_t = e + \frac{1}{2}(u^2 + v^2 + w^2)$$

Using the compact notation $x_i, i=1,2,3$ to represent the x, y, z coordinates respectively and similarly ξ_i for ξ, η, ζ , the shear stress tensor, τ can be written as

$$\tau_{ij} = \mu \left(\frac{\partial \xi_k}{\partial x_j} \frac{\partial u_i}{\partial \xi_k} \right) - \frac{2}{3} \mu \delta_{ij} \frac{\partial \xi_l}{\partial x_k} \frac{\partial u_k}{\partial \xi_l}$$

while

$$b_i = u_j \tau_{ij} + \frac{k}{(\gamma - 1) Pr M_\infty^2} \frac{\partial \xi_l}{\partial x_i} \frac{\partial T}{\partial \xi_l}$$

B. Numerical Setup

We study an unsteady laminar flow over a NACA0012 aerofoil with a sharp trailing edge by DNS. This simulation also provides the base flow field for constructing linear operators. We utilize structured grids with C-type topology for the flow configurations, as shown in figure 1(a). The computational domain has extent $\frac{x}{c} \in [-37.1842, 30.0]$ and $\frac{y}{c} \in [-37.2441, 37.2441]$, which is sufficiently large to capture the unsteady wake and aeroacoustics. The leading edge of the airfoil is position at the origin, at zero angle of attack. The total number of elements over the computational domain is approximately 818×10^3 cells, as shown in figure 1 (a). The height of the first wall cells is set to be $\frac{\Delta y}{c} = 1 \times 10^{-4}$. In the far-field domain, numerical damping is applied to avoid the reflection of outgoing waves. The freestream condition is prescribed at the far-field boundary, whereas the no-slip adiabatic condition is prescribed over the airfoil.

Table 1 Simulation parameters for NACA0012 airfoil

Parameter	Value	Parameter	Value
Simulation grid ($N_x \times N_y$)	1126×727	Filter order (I, J)	(10, 10)
Analysis grid ($N_x \times N_y$)	926×363	Δt (s)	5×10^{-5}
P_{stag} (Pa)	101325	Discretization order	4th explicit
T_{stag} (K)	294.4	Time scheme	RK4
P_{exit} (Pa)	101325	α_{AOA} ($^\circ$)	0.0
T_{wall} (K)	294.4	M_{in}	0.3, 0.5
Re	2×10^5	Boundary conditions	Periodic along k , No-slip on airfoil

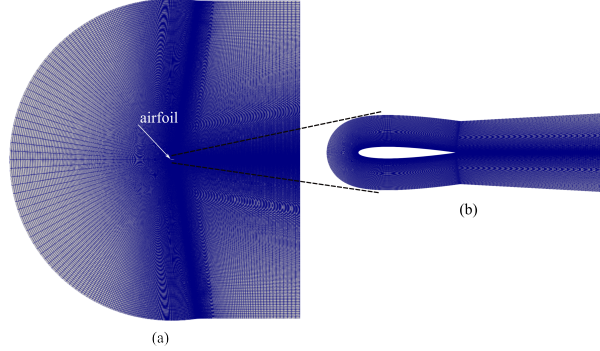


Fig. 1 (a)2D Computational Grid ($N_x \times N_y \equiv (1126 \times 727)$) (b)Analysis Grid($N_x \times N_y \equiv (926 \times 363)$)

IV. Dynamic Mode Decomposition of the flow field

In this section, we use the dynamic mode decomposition(DMD[3]) to extract coherent structures of the flow field. By considering the coherent structures, we extract insights from the flow field and acoustic features associated with dominant frequencies discussed in the previous subsections.

A. Algorithm

We employ the total least squares DMD (tls-dmd [4],[5]). We collect snapshots of the flow field $q_i \in \mathbb{R}^{4N}$ for 400 timesteps with $\Delta t = 0.0125$ between consecutive snapshots, and form two data matrices $\mathbf{X} = [q_0 \dots q_{399}]$, $\mathbf{Y} = [q_1 \dots q_{400}]$, where N is the number of analysis grid cells. The tls-dmd algorithm first performs a dimension reduction of the data matrices $\mathbf{X}$ and $\mathbf{Y}$ with its proper orthogonal decomposition(POD) mode $\psi_r \in \mathbb{R}^{4N \times r}$, where $r \ll 400$ is the number of POD modes selected:

$$X_\psi = \psi_r^T X, \quad Y_\psi = \psi_r^T Y. \quad (2)$$

The POD modes are obtained by performing the singular value decomposition(SVD) of the data matrix: $\mathbf{X} = \psi \Sigma \Phi^T$, where $\psi \in \mathbb{R}^{4N \times 4N}$ and $\Phi \in \mathbb{R}^{m \times m}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{4N \times m}$ is a matrix holding the singular values. The truncated matrices ψ_r , Σ_r and Φ_r are obtained by considering only the first r columns of ψ and Φ , and the first r diagonal elements of Σ . We then compute the SVD of the data matrices, and partition the left singular matrix $U_\psi \in \mathbb{R}^{2r \times 2r}$ into $r \times r$ submatrices, written as

$$\begin{bmatrix} X_\psi \\ Y_\psi \end{bmatrix} = U_\psi \Sigma_\psi V_\psi, \quad U_\psi = \begin{bmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{bmatrix}. \quad (3)$$

The tls-dmd is performed by considering $\tilde{A} = U_{11}U_{12}$. In particular, the DMD eigenmodes η_i and eigenvalues λ_i are given by the eigenvalues and eigenvectors of $\tilde{A}$:

$$\tilde{A}\theta_i = \lambda_i\theta_i, \quad (4)$$

$$\eta_i = \lambda_i^{-1}Y\Phi_r\Sigma_r^{-1}\theta_i. \quad (5)$$

The growth rate γ_i and oscillation frequency in Strouhal number St_i for each DMD mode can be determined from

$$\gamma_i = \frac{\text{Re}(\log(\lambda_i))}{\Delta t_c}, \quad St_i = \frac{\text{Im}(\log(\lambda_i))}{2\pi U_\infty \Delta t_c} \quad (6)$$

Additionally, we employ a compressed sensing method [6] to identify dominant DMD modes from a set of large number of modes, and analyse complex multi-frequencies phenomena such as the trailing-edge noise. This compressed sensing method chooses $s < r$ modes from DMD modes to minimize a reconstruction error defined as

$$\epsilon(S) = \|X_\psi - X_\psi^S\|_2. \quad (7)$$

Here, S is a set of mode numbers that are chosen for reconstruction, and $X_\psi^S \in \mathbb{R}^{n \times m}$ is a reconstructed data matrix from the chosen DMD modes calculated by $X_\psi^S = \Theta^S D$. The matrix $\Theta^S \in \mathbb{C}^{r \times r}$ contains chosen DMD modes for reconstruction and is generated by

$$\Theta^S = [\theta_1 \cdots \theta_r] \begin{pmatrix} \delta_1 & & 0 \\ & \ddots & \\ 0 & & \delta_r \end{pmatrix}, \quad \delta_i = \begin{cases} 1, & \text{if } i \in S, \\ 0, & \text{if } i \notin S. \end{cases} \quad (8)$$

The matrix $D \in \mathbb{C}^{r \times m}$ is the weight matrix to fit the modes to the original data matrix. Here, D is estimated through a least squares regression, that is, $D = \Theta^{S+} X_\psi$, where the superscript $+$ denotes the Moore-penrose pseudo-inverse. With the above-defined reconstruction error function, a greedy mode selection algorithm [7] is applied to find the principal modes for flow reconstruction.

B. Coherent structures of the trailing-edge noise flow

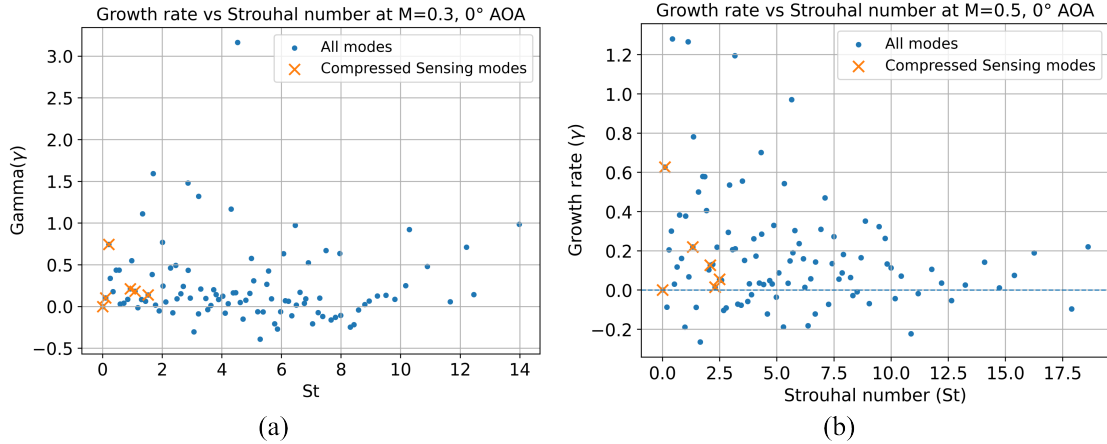


Fig. 2 The DMD spectrum for (a) $M = 0.3$, and (b) $M = 0.5$. The eigenvalues highlighted with orange cross correspond to the selected DMD modes detected by the compressed sensing algorithm.

Figure 2(a) and (b) illustrate top 10 modes chosen by greedy compressed sensing algorithm for $M=0.3$ and $M=0.5$ cases respectively. To validate the compressed sensing algorithm, Power Spectral Density(PSD) analysis is done on both the cases to characterize the tonal frequencies of the trailing edge noise. Figure 3 (a) shows the two probe locations: one upstream(orange) and one downstream(blue) of the airfoil. Figure 3 (a) and (b) shows the PSD of two probe locations for $M=0.5$ and $M=0.3$ cases respectively. All dominant frequencies chosen by compressed sensing algorithm are the characteristics peaks, which serve as the validation. Figure 5 and Figure 4 show the main tonal frequencies responsible for trailing edge noise in $M=0.5$ and $M=0.3$ cases respectively. Figure 5 and 4 (a) represent dipole source which is of low frequency in nature. Figure 5 and 4 (b) represent quadrupole source which is of high frequency. Dipole sound source is generated due to interaction between the vortex and the trailing edge whereas quadrupole source is generated due to interaction between vortices from both sides.

V. Conclusion

We investigated the source of high-frequency quadrupole and low-frequency dipole sound in the trailing edge noise over a 2-D airfoil. Two cases are tested, $M = 0.3$ and $M = 0.5$ with chord based Reynolds number of 2×10^5 and the

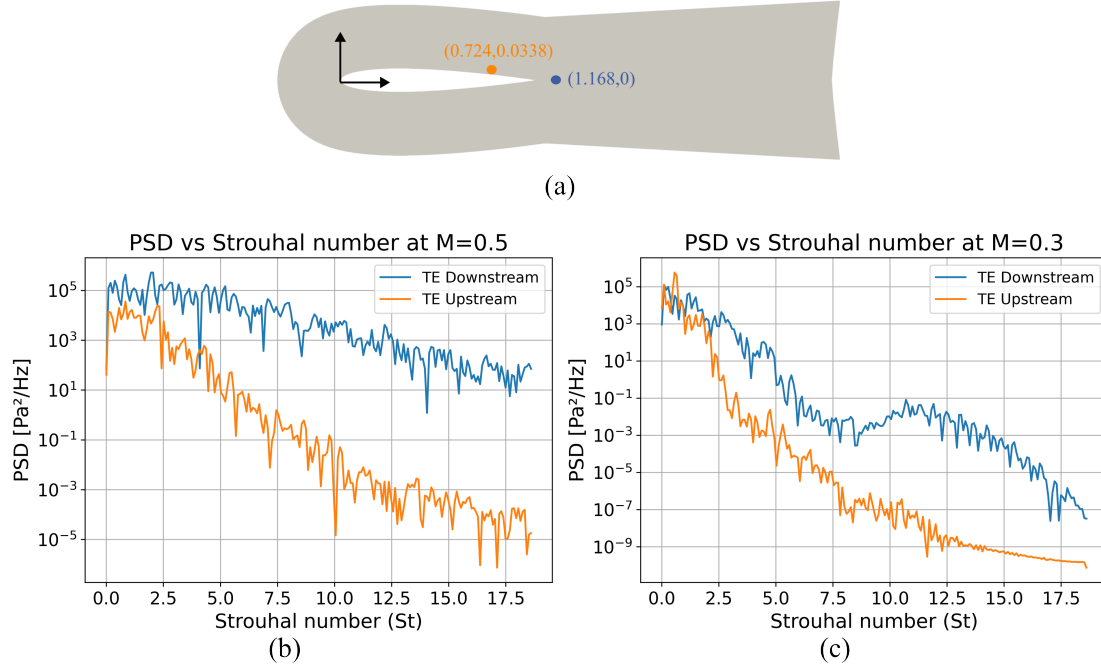


Fig. 3 (a) Probe locations for PSD analysis: one Upstream(orange), one Downstream(blue) of the airfoil (b) PSD analysis for $M = 0.5$ in two probe locations (c) PSD analysis for $M = 0.3$ in two probe locations

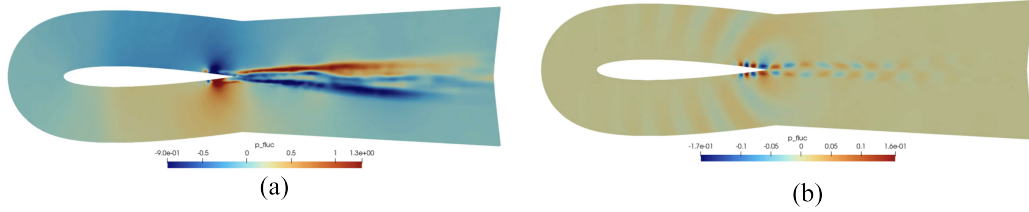


Fig. 4 Pressure fluctuation contour of $M = 0.5$ tonal characteristic modes (a) $St = 0.1$ (b) $St = 2.295$

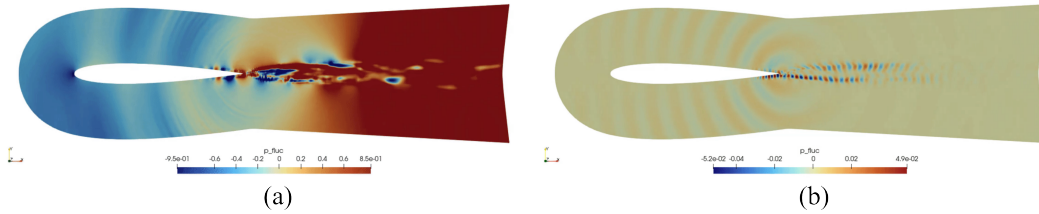


Fig. 5 Pressure fluctuation contour of $M = 0.3$ tonal characteristic modes (a) $St = 0.21$ (b) $St = 1.525$

AOA of 0° . The pressure fluctuation field indicates the intense acoustic waves are radiated from the trailing edge. The frequency spectrum of the pressure fluctuation for $M=0.5$ has a primary packet of peaks around $St = 0.1$ and a higher-frequency secondary packet around $St = 2.295$. The frequency spectrum of the pressure fluctuation for $M=0.3$ has a primary packet of peaks around $St = 0.21$ and a higher-frequency secondary packet around $St = 1.525$. We employed tfs-dmd to evaluate the coherent structures of the flow field. With the use of a compressed sensing algorithm, we identified the modes related to the trailing-edge noise at $St = 0.1$ and 2.295 for $M=0.5$; $St = 0.21$ and $St = 1.525$ for $M=0.3$ case. Dipole tone remains constant in physical frequency as Mach increases, whereas, quadrupole tone shifts to higher physical frequency with increasing Mach due to intensified vortex interactions.

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Code and data - [link](#)

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